

TUBERCULOSIS (TB): DIAGNOSIS

TB tests changing from 'Xpert Ultra' to 'TB NAAT'
(NAAT = nucleic acid amplification test and includes Xpert as well as newer TB tests).

- Check for TB if any TB symptoms: current cough, weight loss, drenching night sweats, fever or coughing up blood.
- Also routinely check for TB, even if no TB symptoms, if:
 - Patient completed TB treatment in last 2 years: check for TB yearly, for 2 years after completing TB treatment.
 - Patient HIV positive: at HIV diagnosis, then yearly (when viral load checked) and if pregnant at first antenatal visit.
- Excluding TB disease during TB preventive treatment (TPT) work up.
- Abnormal TB screening chest x-ray, even if no known TB exposure.

Give urgent attention to the patient with suspected TB and any of:

- Respiratory rate ≥ 30
- Prominent use of breathing muscles
- Coughs up ≥ 1 tablespoon of fresh blood
- Persistent vomiting
- Breathless at rest or while talking
- Drowsy/confused
- Neck stiffness
- New weakness of arm/leg

Manage and refer urgently:

- If breathing difficulty, give face mask oxygen and **ceftriaxone** 1g IV¹/IM to treat for suspected severe pneumonia.
- If able, send 1 sputum sample for TB NAAT. If HIV positive with $CD4 \leq 200$ or WHO clinical stage 3 or 4, also do a rapid urine LF-LAM test.

Start the workup to diagnose TB in the patient not needing urgent attention

Test sputum

- Send 1 sputum sample for TB NAAT: demonstrate how to give sputum sample $\rightarrow 91$.
- If unable to produce sputum: induce sputum $\rightarrow 91$. If unsuccessful, arrange chest x-ray $\rightarrow 93$.
- Ask patient to return for results in 2 days. Manage according to results: if TB NAAT positive (or trace), manage below. If TB NAAT negative $\rightarrow 93$.

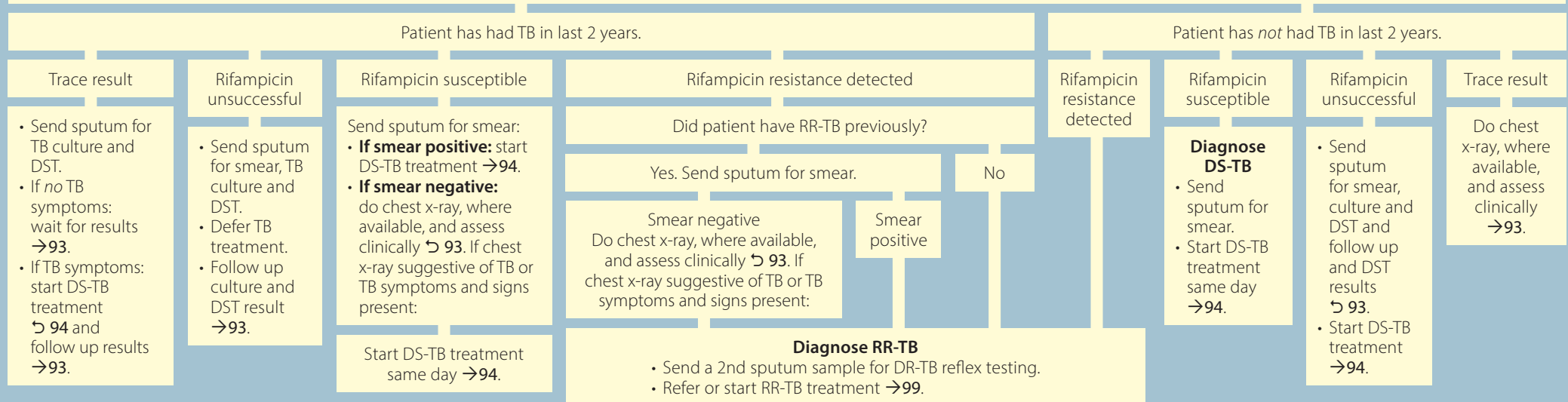
Test blood

Test for HIV
 $\rightarrow 110$.

Test urine

- If HIV positive and $CD4 \leq 200$ or WHO stage 3 or 4 disease, also do rapid urine LAM test (only if TB symptoms):
 - If LAM positive, **diagnose TB** and start DS-TB treatment same day $\rightarrow 94$ and follow up TB NAAT result (see below).
 - If LAM negative, ask patient to return for TB NAAT results in 2 days.

TB NAAT positive or trace (MTB detected)



¹Do not mix Ringer's lactate and IV ceftriaxone. Flush IV line with **sodium chloride 0.9%** before and after IV ceftriaxone.